

Smart Animal Detector System using Arduino Nano and PIR Sensor

Abstract

This project implements a low-cost smart animal detector for farms and wildlife-sensitive zones. A PIR sensor detects movement, and an Arduino Nano activates LED and buzzer alerts. The system is simple, reliable, beginner-friendly, and suitable for academic submission.

1. Introduction

Unwanted animal movement near crops or restricted areas can cause losses. A compact alerting system helps users respond quickly. This project demonstrates an embedded motion-alert design using minimal components.

2. Objectives

- Detect movement using PIR sensor
- Trigger immediate LED and buzzer alerts
- Keep implementation simple for students and beginners
- Provide complete report and test workflow for evaluation

3. Hardware Requirements

Refer to: docs/components-list.md

4. Software Requirements

- Arduino IDE 1.8+ or 2.x
- USB driver and cable for Arduino Nano
- Serial Monitor at 9600 baud for runtime logs

5. Circuit Design and Connections

Refer to: docs/circuit-connections.md

6. Source Code

Refer to: arduino/SmartAnimalDetector/SmartAnimalDetector.ino

7. Working Explanation

Refer to: docs/working-principle.md

8. Testing and Validation

Refer to: docs/testing-procedure.md

9. Applications

- Farm boundary monitoring
- Night-time movement alerts
- Wildlife trail notification points
- Embedded systems lab demonstrations

10. Advantages

- Low cost
- Easy implementation
- Real-time alerting
- Expandable to IoT (GSM/Wi-Fi notifications)

11. Limitations

- Detects movement only (no animal type classification)
- Environmental temperature and placement affect sensitivity

12. Future Scope

- Add GSM module for SMS alerts
- Add camera trigger for evidence capture
- Add solar power and battery backup
- Integrate cloud dashboard for remote monitoring

Conclusion

The Smart Animal Detector System is a practical and effective starter project for IoT and embedded learning. With complete code, connections, and test procedure, it is ready for academic submission.